

Lecture Notes on Numerical Methods

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RECAP

Key points from the previous class:

- ▶ Point 1: Exhaustive Search Method and order of Convergence.
- ▶ Point 2: Bisection Method(Interval Halving) and its Order of Convergence.
- ▶ Point 3: Regula - Falsi Method and its Order of Convergence.

Order of Convergence of Root-Finding Methods

Method	Order of Convergence	Notes
Bisection	$p = 1$ (linear)	Guaranteed, slow
Regula Falsi	$p = 1$ (linear)	Can be faster than Bisection in practice, but still linear; convergence can be slow if one endpoint remains fixed

Fixed Point Iteration

Basic definition and theroms :

Definition(Fixed point:)

A real number p is called a **fixed point** of g if $g(p) = p$.

Example:

$x = 0$ is fixed point for $\sin x$ because $\sin 0 = 0$

Example (Fixed point and root)

Suppose $\alpha \neq 0$. Show that p is a root of $f(x)$ iff p is a fixed point of

$$g(x) := x - \alpha f(x)$$

Fixed Point Theorem

Theorem:

1. If $g \in C[a, b]$ and $a \leq g(x) \leq b$ for all $x \in [a, b]$, then g has at least one fixed point in $[a, b]$.
2. If, in addition, g' exists in $[a, b]$, and $\exists k < 1$ such that $|g'(x)| \leq k < 1$ for all x , then g has a unique fixed point in $[a, b]$.

Example

Example:

Let $f(x) = \cos(x)$ on $[0, 1]$.

- ▶ $f(x)$ is continuous on $[0, 1]$.
- ▶ Since $\cos(x) \in [0, 1]$ for $x \in [0, 1]$, $f : [0, 1] \rightarrow [0, 1]$.
- ▶ By the fixed-point theorem, $f(x) = \cos(x)$ has at least one fixed point in $[0, 1]$.

Numerically solving $\cos(x) = x$ gives a fixed point near $x \approx 0.739$.

Example (Fixed Point Theorem – Failed Case 1)

$g(x) = \frac{x^2 - 1}{3}$ has a unique fixed point in $[3, 4]$.

But we can't use FPT to show this.

- ▶ Note that there is a unique fixed point in $[3, 4]$ ($p = \frac{3+\sqrt{13}}{2}$), but $g(4) = 5 \notin [3, 4]$, and $g'(4) = \frac{8}{3} > 1$ so we cannot apply FPT here.
- ▶ From this example, we know FPT provides a *sufficient* but not *necessary* condition.

Fixed Point Iteration

Theorem: Let $g : [a, b] \rightarrow [a, b]$ be a function such that

- (i) $g(x)$ is continuous on $[a, b]$
- (ii) $g(x)$ is differentiable on (a, b)
- (iii) $\exists k \in (0, 1)$ such that $|g'(x)| \leq k < 1, \forall x \in (a, b)$

Then,

1. Sequence $\{p_n\} \rightarrow p$, where $p_n = g(p_{n-1})$ for any $p_0 \in [a, b]$;
2. $|p_n - p_{n-1}| \leq k^n \max(p_0 - a, b - p_0)$; and,
3. $|p_n - p| \leq \frac{k^n}{1-k} |p_1 - p_0|$.

Remark 1: Interval $[a, b]$ must be closed and bounded.

Remark 2: All three given conditions guarantee that there exist one and only one fixed point p in $[a, b]$.

Proof contd.

Importance of k : When k is small, the error decreases quickly, so order of convergence becomes higher but as k approaches 1, the order of convergence becomes much slower. □

Example

Consider the function $f(x) = x^3 + x^2 - 3x - 3$. Now we have to find the zero of $f(x)$ in $(1, 2)$. ($\because f(1)f(2) = -4 \cdot 3 < 0$)
ie

$$f(x) = 0$$

$$x^3 + x^2 - 3x - 3 = 0$$

$$\frac{x^3 + x^2 - 3}{3} = x$$

$$g(x) = x$$

So, here zero of $f(x)$ and fixed point of $g(x)$ are same.

Remark: Every rootfinding problem can be transformed into multiple fixed point problems. Some of these fixed point problems converge rapidly, some will converge slowly and some will not converge at all.

Example contd.

$f(x) = x^3 + x^2 - 3x - 3 = 0$ can be written as following fixed point problems.

$$g_1(x) = \frac{1}{3}(x^3 + x^2 - 3)$$

$$g_2(x) = -1 + \frac{3x + 3}{x^2}$$

$$g_3(x) = \sqrt[3]{3 + 3x - x^2}$$

$$g_4(x) = \sqrt{\frac{3 + 3x - x^2}{x}}$$

$$g_5(x) = x - \frac{x^3 + x^2 - 3x - 3}{3x^2 + 2x - 3}.$$

Now, we are going to check which of the above problems converge or diverges, generated by sequence $p_n = g(p_{n-1})$ and $p_0 = 1$.

Example contd.

n	$g_1(x)$	$g_2(x)$	$g_3(x)$
0	+1.0	+1.0	+1.0
1	-0.3333333333	5.0000000000	1.709975947
2	-0.9753086420	-0.2800000000	1.733134316
3	-0.9921709716	26.5510204100	1.731994802
4	-0.9974310264	-0.8827544117	1.732053695
5	-0.9991480696	-0.5486245114	1.732050659
6	-0.9997165068	3.4989256110	1.732050815
7	-0.9999055558	0.1024544340	1.732050807
8	-0.9999685246	314.0796731	-
9	-0.9999895088	-0.9904178717	-
10	-0.9999965030	-0.9706946913	-

Table 1: $g_1(x)$ converges to $-1 \notin (1, 2)$, $g_2(x)$ diverges, and $g_3(x)$ converges rapidly.

Example contd.

n	$g_4(x)$	$g_5(x)$
0	+1.0	+1.0
1	2.236067978	-3.000000000
2	1.451059202	-2.200000000
3	1.901682432	1.830150754
4	1.635808067	1.737795453
5	1.788336635	1.732072292
6	1.699764653	1.732050808
7	1.750767137	1.732050808
8	1.721269132	—
9	1.738283891	—
10	1.738454854	—

Table 2: $g_4(x)$ converge slowly while $g_5(x)$ converges rapidly

Question: How to know which g_i converges to its fixed point ?

Order Of convergence for FPI

Theorem: Let $g : [a, b] \rightarrow [a, b]$ be a function such that

- (i) $g(x)$ is continuous on $[a, b]$.
- (ii) $g'(x)$ is continuous on (a, b) .
- (iii) $|g'(x)| \leq k < 1, \forall x \in (a, b)$

Claim: If $g'(p) \neq 0$, then for any $p_0 \in [a, b]$ the sequence generated by $p_n = g(p_{n-1})$ **converges linearly** to the fixed point p .

Example 1:

The equation $x^3 - 4x - 9 = 0$ has a root in the interval $(2, 3)$. Use Fixed-Point method to show that the iteration function

$$\phi(x) = \frac{x^3 - 9}{4}$$

generates a divergent sequence, while the iteration function

$$\phi(x) = \frac{x^3 - 19x - 9}{15}$$

generates a convergent sequence. Start with any initial approximation in the interval $(2, 3)$.

Solution:

From the given equation, $x^3 - 4x - 9 = 0$ we have following iteration function

$$\phi(x) = \frac{x^3 - 9}{4}$$

The root lies in $(2, 3)$. Choose $x_0 = 2.7$.

$$x_1 = \phi(x_0) = 2.670750$$

$$x_2 = \phi(x_1) = 2.512554$$

Further iterations:

x_n	$\phi(x_{n-1})$
x_3	1.715393
x_4	-0.988084
x_5	-2.491169
x_6	-6.114999
x_7	-59.414871
x_8	-52437.757812
x_9	$-3.6047266250752 \times 10^{12}$

Hence the sequence diverges to $-\infty$.

Solution:

From the given equation, $x^3 - 4x - 9 = 0$ we take the iteration function

$$\phi(x) = \frac{x^3 - 19x - 9}{15}.$$

The root lies in $(2, 3)$. Choose $x_0 = 2.5$.

$$x_1 = \phi(x_0) = 2.722500,$$

$$x_2 = \phi(x_1) = 2.702678$$

Further iterations:

x_n	$\phi(x_{n-1})$
x_3	2.707284
x_4	2.706378
x_5	2.706558
x_6	2.706522
x_7	2.706529
x_8	2.706528
x_9	2.706528

Hence the sequence converges to

$$p \approx 2.7065.$$

Example: 2

Consider the function $g(x) = e^{-x^2}$.

- (a) Prove that g has a unique fixed point on the interval $[0, 1]$.
- (b) With a starting approximation $p_0 = 0$, use the iteration scheme

$$p_n = e^{-p_{n-1}^2}$$

to approximate the fixed point on $[0, 1]$ to within 5×10^{-7} .

- (c) Use the theoretical error bound

$$|p_n - p| \leq \frac{k^n}{1 - k} |p_1 - p_0|$$

to obtain a theoretical bound on the number of iterations needed to approximate the fixed point to within 5×10^{-7} . How does the number of iterations performed in part (b) compare with the theoretical bound?

Part (a): Existence of Fixed Point

Given

$$g(x) = e^{-x^2}, \quad x \in [0, 1].$$

We apply the Fixed Point Theorem on $[0, 1]$.

Step 1: Range of $g : [0, 1] \rightarrow [0, 1]$.

For $x \in [0, 1]$, we have $0 \leq x^2 \leq 1$, so

$$e^{-1} \leq e^{-x^2} \leq 1.$$

Hence

$$g(x) = e^{-x^2} \in [e^{-1}, 1] \subset [0, 1].$$

Step 2: Continuity.

The function $g(x) = e^{-x^2}$ is continuous on $[0, 1]$. Therefore, by the Fixed Point Theorem, g has at least one fixed point in $[0, 1]$.

Part (a): Uniqueness of Fixed Point

Step 3: Derivative bound.

$$g'(x) = -2xe^{-x^2}, \quad |g'(x)| = 2xe^{-x^2}.$$

Let $h(x) = 2xe^{-x^2}$. Then

$$h'(x) = 2e^{-x^2}(1 - 2x^2),$$

which is zero at $x = \frac{1}{\sqrt{2}}$. Thus

$$\max_{[0,1]} |g'(x)| = 2 \cdot \frac{1}{\sqrt{2}} e^{-1/2} = \sqrt{2} e^{-1/2} \approx 0.8578 < 1.$$

Hence there exists $k < 1$ such that

$$|g'(x)| \leq k \quad \text{for all } x \in (0, 1).$$

By the Fixed Point Theorem, the fixed point of g in $[0, 1]$ is unique.

$g(x) = e^{-x^2}$ has a unique fixed point in $[0, 1]$.

Part (b): Fixed Point Iteration

We use the iteration scheme

$$p_{n+1} = e^{-p_n^2}, \quad p_0 = 0.$$

Starting from $p_0 = 0$, the sequence $\{p_n\}$ is generated by repeatedly evaluating $p_{n+1} = e^{-p_n^2}$.

We continue the iteration until the stopping criterion

$$|p_{n+1} - p_n| < 5 \times 10^{-7}$$

is satisfied. Using a computer, this occurs after 89 iterations, and the corresponding approximation of the fixed point is

$$p \approx 0.6529188603.$$

$p \approx 0.6529188603$

Fixed Point Iteration for $p = e^{-p^2}$

n	p_n	Error = $ p_{n+1} - p_n $
0	0.0000000000	NaN
1	1.0000000000	1.0000000000
2	0.3678794412	0.6321205588
3	0.8734230185	0.5055435773
4	0.4663271888	0.4070958296
5	0.8045589442	0.3382317554
$\vdots$	$\vdots$	$\vdots$
86	0.6529182857	0.0000007708
87	0.6529189429	0.0000006572
88	0.6529183825	0.0000005603
89	0.6529188603	0.0000004778

Table 3: Iteration values and error

Part (c): Error Bound

From Part (a),

$$k = \max_{[0,1]} |g'(x)| = \sqrt{2} e^{-1/2} \approx 0.8578 < 1.$$

The Fixed Point Theorem gives

$$|p_n - p| \leq \frac{k^n}{1 - k} |p_1 - p_0|.$$

Since $p_0 = 0$ and $p_1 = 1$, we get

$$|p_n - p| \leq \frac{k^n}{1 - k}.$$

We require

$$|p_n - p| \leq 5 \times 10^{-7}.$$

Thus,

$$\frac{k^n}{1 - k} \leq 5 \times 10^{-7}.$$

Since $1 - k = 0.1422361$, this gives

$$k^n \leq 7.11 \times 10^{-8}.$$

Part (c): Final Estimate

Taking logarithms,

$$n \geq \frac{\ln(7.11 \times 10^{-8})}{\ln(0.8578)} \approx 107.4.$$

Hence

$$\boxed{n \geq 108}.$$

From Part (b), the actual number of iterations required was

89.

Thus the theoretical bound of iteration is larger than the actual count, as expected.

Example: 3

Use the Fixed-Point method to compute the root of the equation $x - e^{-x} = 0$ in the interval $x \in (0, 1)$.

Solution:

Example: 3

Use the Fixed-Point method to compute the root of the equation $x - e^{-x} = 0$ in the interval $x \in (0, 1)$.

Solution: Rewrite the equation as

$$x = e^{-x}.$$

We choose the iteration function

$$\phi(x) = e^{-x}.$$

Since $f(0) = -1$ and $f(1) = 0.631$, the root lies in $(0, 1)$. Choose $x_0 = 0.5$.

$$\begin{aligned}x_1 &= e^{-0.5} = 0.606531, \\x_2 &= e^{-0.606531} = 0.545239.\end{aligned}$$

Further iterations are given below.

Successive Approximations

x_n	Value	x_n	Value
x_1	0.606531	x_{10}	0.566907
x_2	0.545239	x_{11}	0.567277
x_3	0.579703	x_{12}	0.567067
x_4	0.560065	x_{13}	0.567186
x_5	0.571172	x_{14}	0.567119
x_6	0.564863	x_{15}	0.567157
x_7	0.568438	x_{16}	0.567135
x_8	0.566409	x_{17}	0.567148
x_9	0.567560	x_{18}	0.567141

Hence the root is $p \approx 0.567141$

Graphical iteration of $y = x$ and $y = e^{-x}$

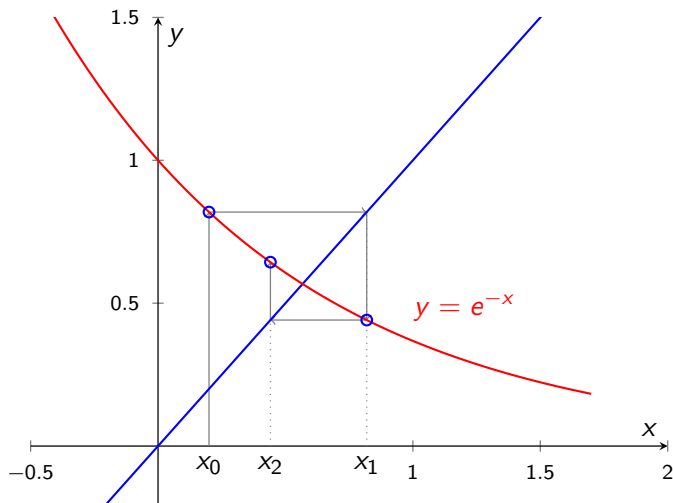


Figure 1: Visualization of Fixed Point Method.

Fixed Point Iteration

Exercise:

Verify that $x = 1/a$ is a fixed point of the function

$$g(x) = x(2 - ax).$$

Use the techniques of this section to determine the order of convergence and the asymptotic error constant of the sequence

$$p_n = g(p_{n-1})$$

toward $x = 1/a$.

Ans:

$$\text{Fixed point } \alpha = \frac{1}{a}$$

$$\text{Order of convergence} = 2$$

$$\text{Asymptotic error constant} = |a|$$